Numerical Optimisation and Model Predictive Control
A Hands-On Journey with CasADi and Impact
CCTA 2026

M. Bos, L. Callens, W. Decré, A. Flórez, J. Gillis, E. Sandra, M. Schietecat, J. Swevers

August 11, 2026

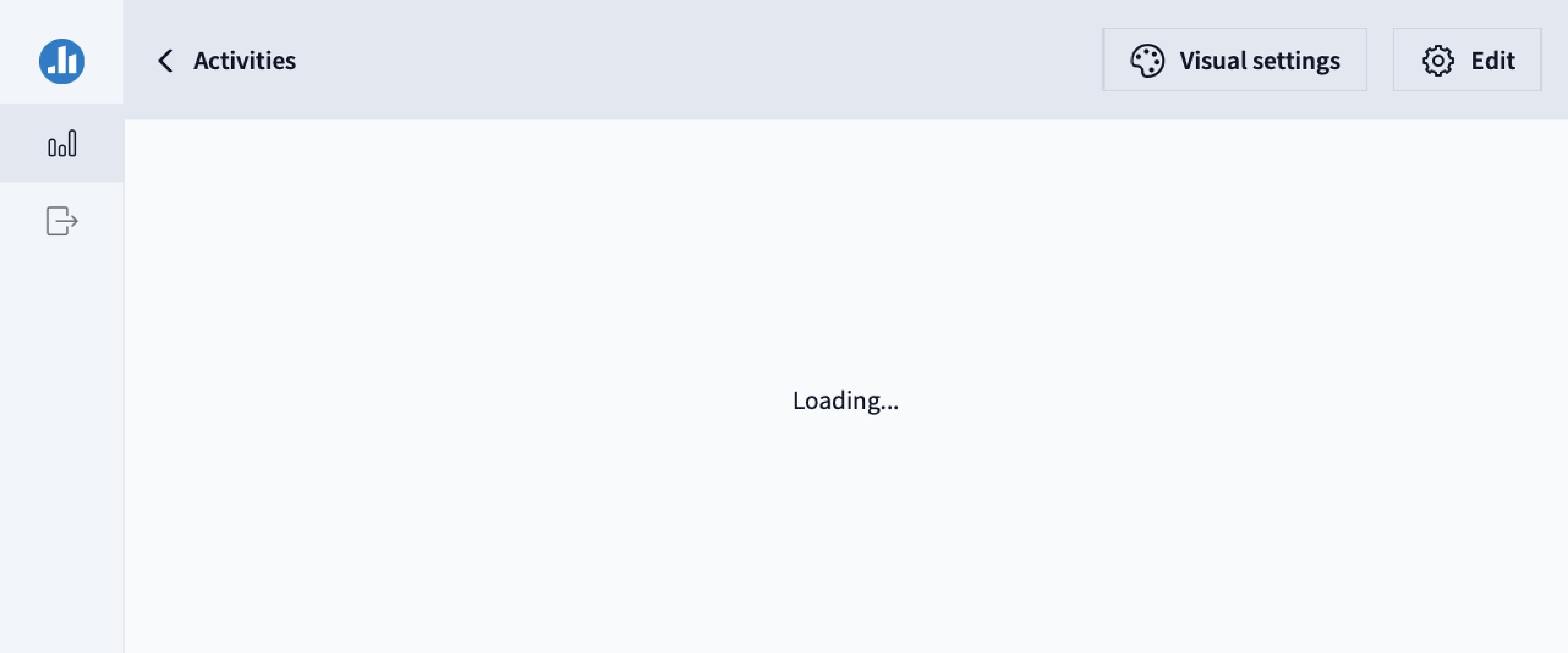
Software overview

- All released as free and open-source software

CasADi	Rockit	Impact	ImpaC++t
• self-contained symbolic framework for nonlinear programming • algorithmic differentiation • interfaces to own & third-party solvers • C-code generation	• optimal control prototyping • free end-time & multi-stage problems	• from optimal control prototyping to NMPC deployment • automatic generation of deployable artifacts	• closed-loop control in C++ • real-time scheduling & I/O • simulation and deployment • Linux RT kernel

- Additional software: Fatrop solver

Poll



Workshop overview

Session	Start	Topic
Le. 1	8:30 a.m.	Welcome & workshop overview Software installation Lecture CasADi – part 1
Br. 1	10:00 a.m.	Break
Le. 2	10:30 a.m.	Lecture CasADi – part 2 Exercise Alvik – static optimisation
Br. 2	12:00 p.m.	Lunch break
Le. 3	1:30 p.m.	Lecture MPC with Impact Exercise Alvik – dynamic optimisation
Br. 3	3:00 p.m.	Break
Le. 4	3:30 p.m.	Lecture ImpaC++t Exercise Furuta pendulum

KU LEUVEN

Software installation

- Available at https://github.com/meco-group/CCTA-impact-2026#software-installation

SCAN ME

CCTA-CasADi-Impact

Questions?